

The liver is zoned — and a paper says disease un-zones it

PERICENTRAL zone

detox / drug metabolism

PERIPORTAL zone

urea cycle

WHAT WAS KNOWN · zonation atlas

A healthy-liver atlas mapped the **marker genes** that say which zone a cell belongs to. That is our **ruler** for “what zonation is.”

THE CLAIM · Gribben et al., Nature 2024

In fatty-liver disease (MASLD), hepatocytes:

1. **lose their zonation** (the two groups blur together), and
2. **turn into bile-duct cells** (transdifferentiation).

OUR IDEA

Take the atlas’s healthy-zonation ruler, and measure the “gap” across the MASLD disease stages — does it really shrink?

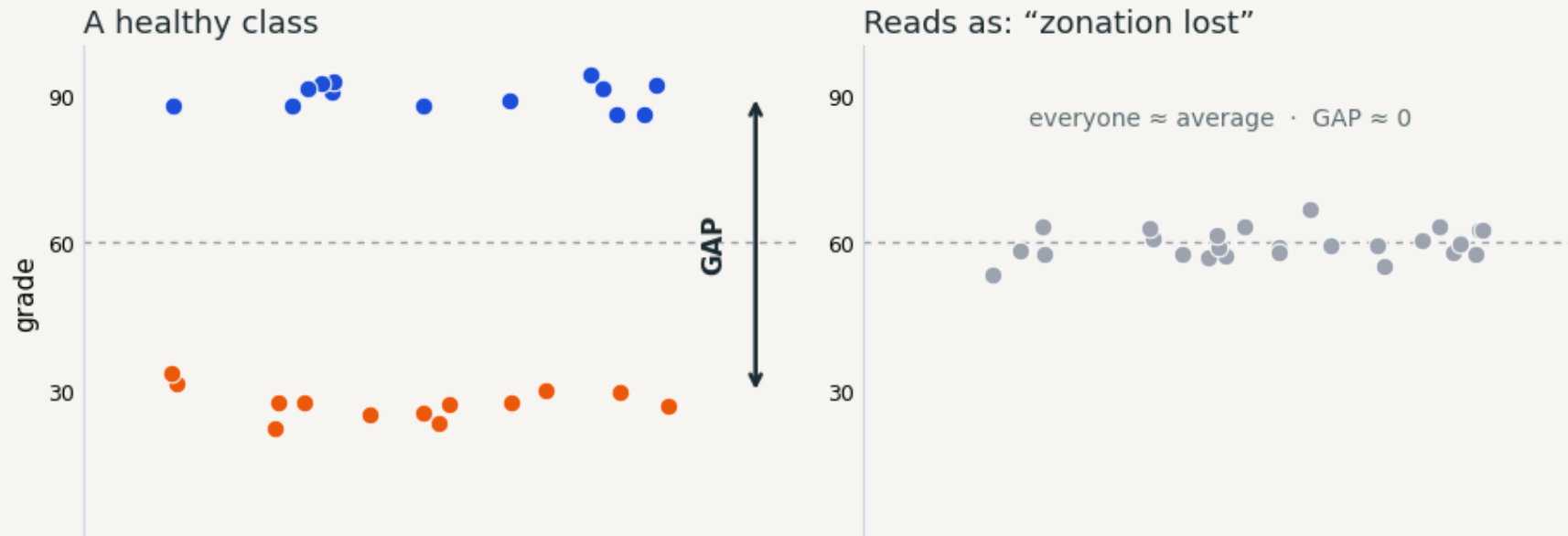
Our question

Does hepatocyte zonation really degrade across MASLD — and is it linked to the hepatocyte→bile-duct transdifferentiation the paper describes?

We re-checked both claims in the raw data — and ran into two catches: first our ruler, then the experiment itself.

The first ruler — a school-class analogy

Imagine a make-believe school where — by design — every student is clearly one of two kinds: **top** (≥ 90) or **failing** (≤ 30), *nobody in between*. (A thought experiment — real grades vary, but the two-bin picture makes the idea clear.)

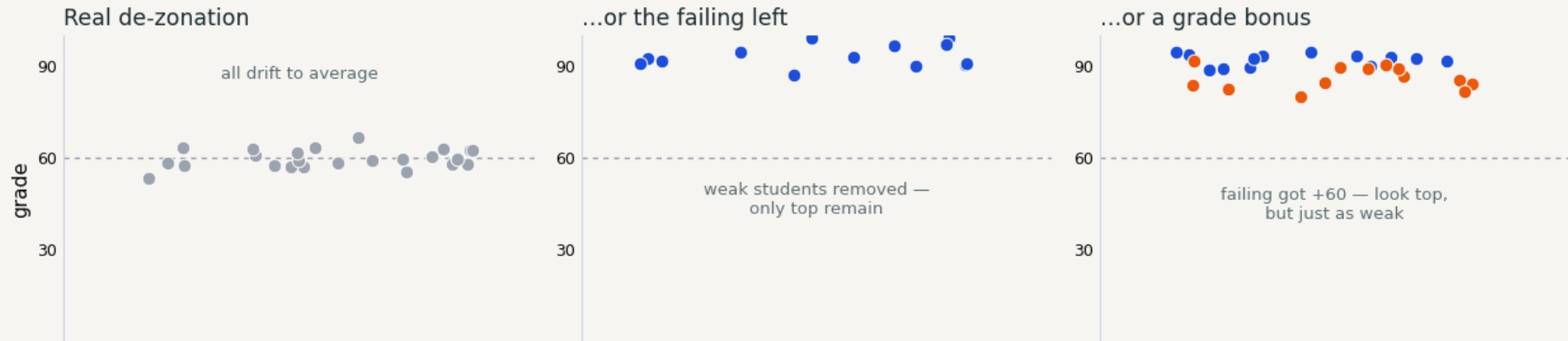


In this imagined class, two clear groups (top = pericentral cells, failing = periportal). The first method measures only the GAP between them.

The naive reading: if the gap collapses to zero — everyone looks average — it reads as “zonation lost.”

A zero gap can mean very different things

Each gives $GAP \approx 0$ — the relative ruler can't tell them apart



Three classes that all read $GAP \approx 0$ — yet only the first is real de-zonation.

It hides who moved

Real de-zonation? A top $\leftrightarrow$ failing swap? Or did the failing students simply leave? The gap reads the same.

It's bent by tweaks

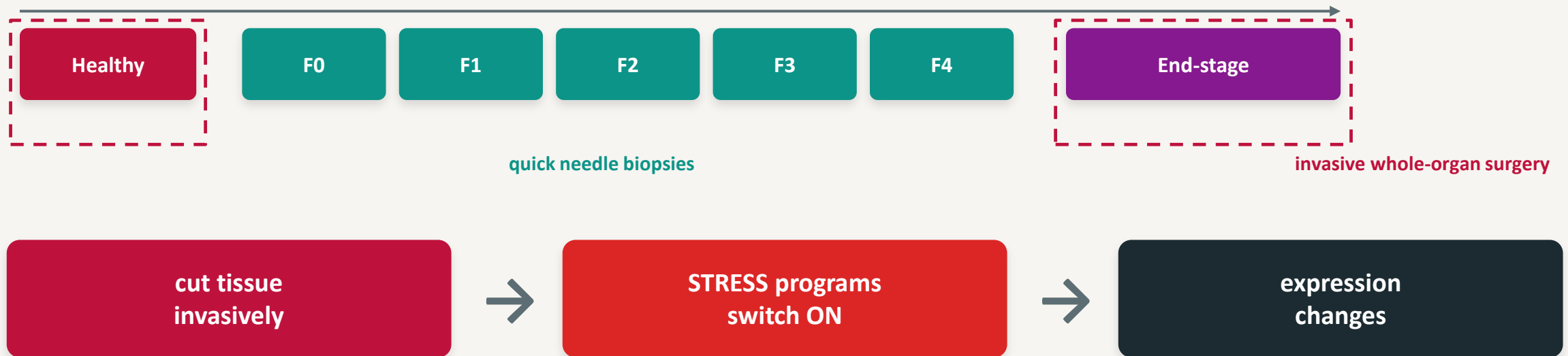
Give the failing a grade bonus and they "look" top: gap = 0, yet nothing real changed. Any across-the-board shift fools a relative ruler.

One number, re-graded data

A single relative score crams all of these together — and it's built on normalized grades tuned for the original question, not ours.

→ So we drop the gap and count the raw work each cell does — actual molecules, not a relative score — which can tell these apart.

Could the setup itself fake the signal?



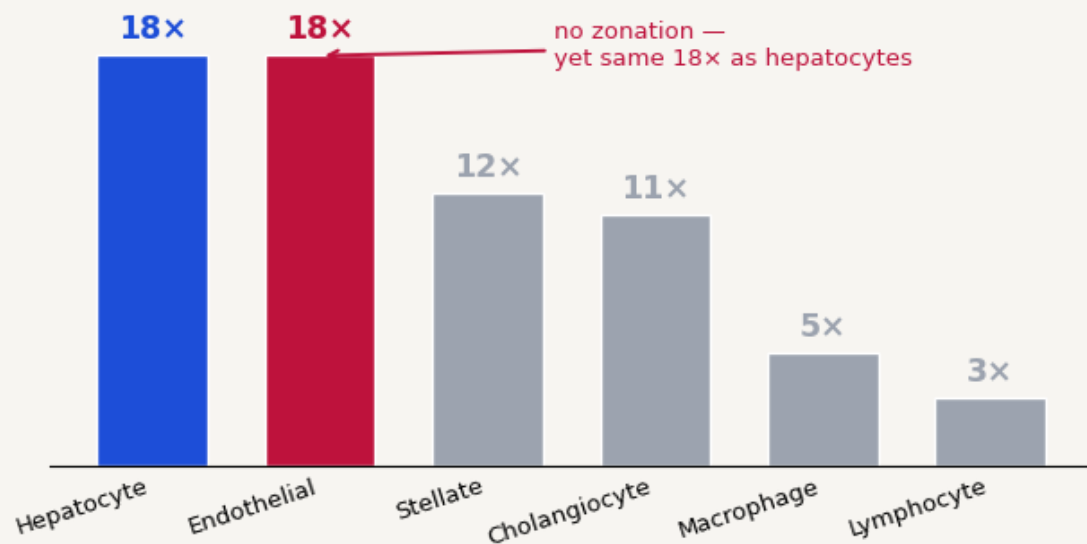
So the two ends carry an extra stress signal.

Is it coming from how they were collected — and could it be faking the “de-zonation”?

We ran two checks →

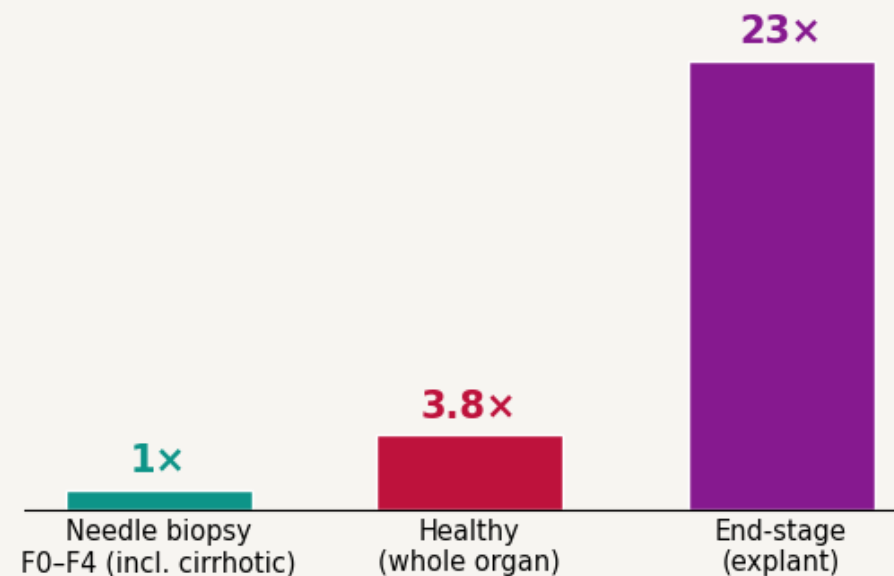
Could the stress be the real signal? Two reasons to doubt it

① Can't be pinned on zonation



Even if the stress *is* a zonation change, we can't separate it from handling — cells with **no zonation program at all** (e.g. endothelium) show the same spike. So it's at least confounded.

② Doesn't behave like disease



If it were the disease, this pattern is odd: **loud in HEALTHY tissue** and **silent in cirrhosis**. A disease signal should rise with severity — so the pattern leans to procurement.

→ so we look at every group — healthy · F0 · F1–F4 · end-stage — but the two ends are procurement-confounded; disease is read from the matched biopsies (F1–F4).

Our fix: count, control, look finer

The two catches told us what to fix. So we changed three things — same data, read more honestly:

1

Count real molecules

We stop measuring the relative gap and count the actual molecules in each cell — so we can see who moved and how many.

fixes: the gap hid who moved

2

Control the grading

We pin down the normalization ourselves, so an across-the-board tweak — an artifact — can't quietly shift the answer.

fixes: the gap was bent by tweaks

3

Look finer than two bins

Not just top vs failing. For every cell: clearly pericentral? periportal? both? neither? — and how loud is its program?

fixes: the coarse top / failing split

Same data — raw, normalization-controlled, and read at a finer grain. Now the ruler can't fool us.